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يُونِيسُيْتِيْ اِسْلَامْ اِنْتَارَا بَغْسِيَا مِلِّيْسِيَا
Garden of Knowledge and Virtue

MCTA 3203

MECHATRONICS SYSTEM INTEGRATION

SEMESTER 1, 25/26

CONCEPTUAL DESIGN DETAILS

SECTION 2

GROUP 19

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1.0 SYSTEM ARCHITECTURE

Perception and Input Layer

The system begins its process by gathering environmental data through specialized sensors and user inputs. A PIXYCAM is utilized to scan the RGB values of the laundry to determine if the load consists of white or colored cloth. This data is essential for the system to automate the washing parameters later in the cycle. Additionally, a lid sensor acts as a safety check to ensure the machine is properly closed before any mechanical operation begins. The user provides the final input via a physical button to select the specific operational mode, such as Wash, Rinse, or Spin.

Control and Data Layer

The control layer serves as the logic center that processes sensor data and makes operational decisions. Once the PIXYCAM identifies the cloth type, the system stores this data to be retrieved during the washing phase. If the lid sensor returns a "No" status, the control logic triggers a safety loop that displays a warning and halts progress until the lid is secured. When the Wash mode is activated, the system automatically retrieves the stored color data to set specific profiles: "White Cloth" triggers a gentle speed with a longer duration, while "Color Cloth" triggers a normal speed with standard duration.

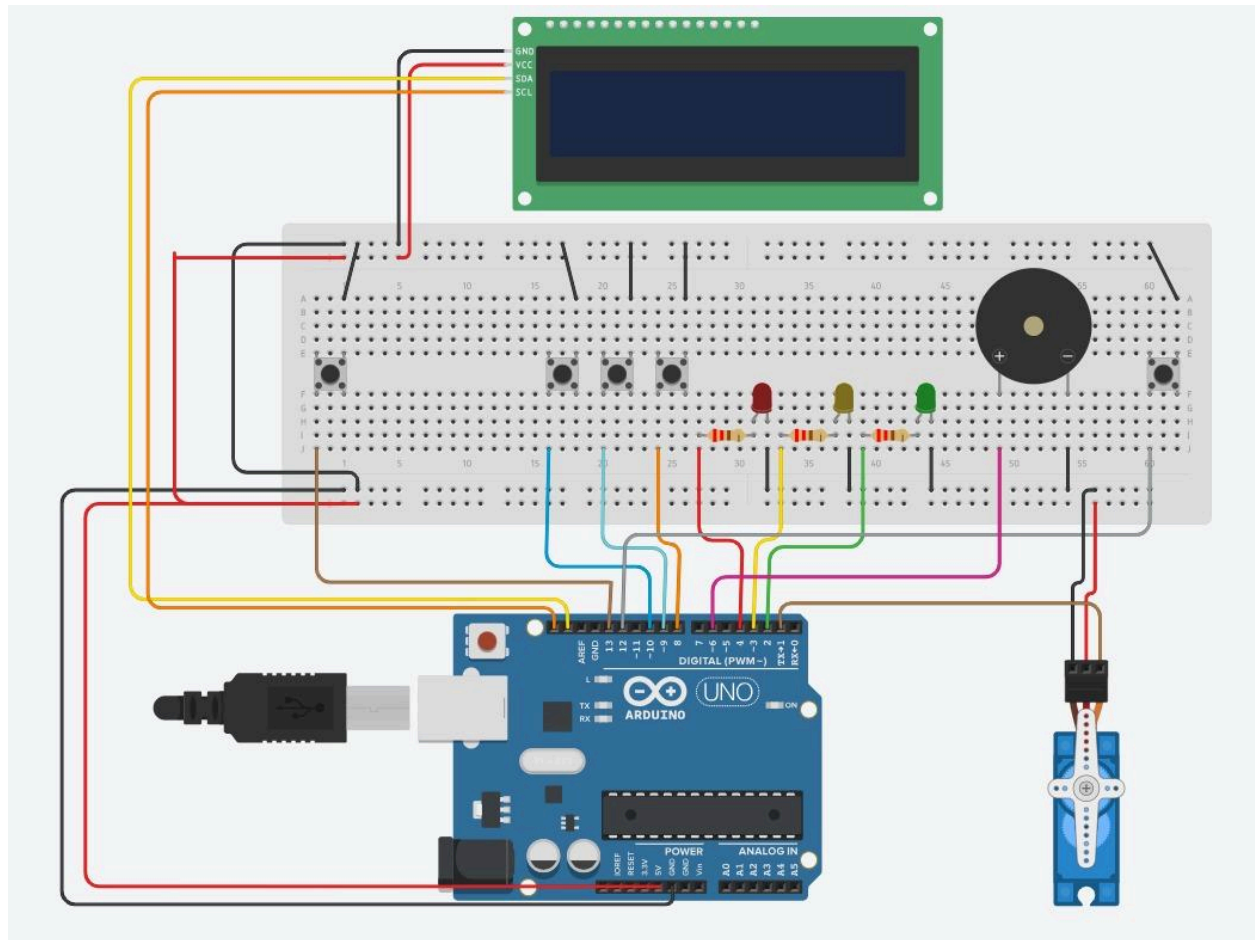
Interaction and Feedback Layer

This layer manages the communication between the machine and the user through visual and auditory signals. The LCD display provides real-time status updates, such as the current mode, time taken, and instructional prompts like "Please Close The Lid". Auditory feedback is provided by a buzzer that beeps once for white cloth and twice for color cloth to confirm successful scanning. Furthermore, the system includes a visual monitoring safety feature using LEDs: a Green LED illuminates if the drum is rotating correctly, while a Red LED warns the user if the drum has failed to rotate.

Actuation and Hardware Layer

The actuation layer consists of the physical components that perform the mechanical work of the laundry cycle. The drum motor is the primary actuator, capable of rotating in both Clockwise (CW) and Counter-Clockwise (CCW) directions during Rinse and Wash cycles to agitate the clothes. For the Spin mode, the motor switches to a high-speed Clockwise rotation to extract water. After these cycles are completed, the system activates the drain mechanism, signaled by the LCD display, to remove used water before reaching the final "Stop" state.

2.0 WIRING DIAGRAM



3.0 FLOW CHART

